Extension of the Hypothesis: Gravity as a Consequence of Energy Distribution

A Lay Hypothesis

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Date: October 17, 2026

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1. Starting Point

The fundamental assumption of the hypothesis reads:

Universe = E

The universe is understood as a connected physical energy system in which matter does not represent a fundamentally independent category, but is conceived as a specific bound and organized state of energy.

On this foundation, the further question arises whether gravity might also be interpreted differently than in the standard geometric description of General Relativity (GR).

The present extension does not claim that GR has been refuted. Rather, it explores the speculative possibility that observed gravitational effects could be based on a more fundamental physical effect of the energy distribution.

2. Energy Distribution and Acceleration

In the standard description of gravity, the motion of bodies is explained through the geometry of spacetime. The present consideration raises the question of whether this geometric description might not represent the most fundamental explanation of the underlying physical process.

Instead, it is assumed that the spatial distribution of a total system's energy could produce a physical effect through which other bodies are accelerated.

Using Earth as an example, this would mean that while the center of the Earth coincides with the system's center of mass, the central significance of this point might rest on a deeper-lying distribution of energy. Acceleration toward the center of the Earth would accordingly not necessarily be understood as the result of a fundamentally curved spacetime, but could emerge from the effect of the energy distribution of the overall system.

In this context, it would need to be investigated how the strength of this effect depends on total energy, its spatial distribution, and the distance to the center of the system.

3. Universal Acceleration

An essential starting point of this consideration is the observation that bodies of different masses accelerate at approximately the same rate under comparable conditions when other influences, such as air resistance, are neglected.

Within the present hypothesis, this could be interpreted as meaning that the decisive acceleration is determined by the energy distribution of the causing overall system, rather than primarily by the mass or internal composition of the accelerated body.

Earth would thus set the conditions for the motion of a falling body. The properties of the body itself would not determine the acceleration in the same way that might occur in a conventional interaction between two objects of different masses.

This interpretation is not yet mathematically elaborated. In particular, it would need to be clarified how such a description differs quantitatively from GR and under what conditions both yield the same results.

4. The Potential Role of Electromagnetic Fields

Another open question is whether the assumed effect of the energy distribution could be related to electromagnetic fields.

Matter consists of systems of electrically charged components, even though many macroscopic bodies are largely electrically neutral to the outside world. However, this neutrality does not mean that no electromagnetic interactions occur within such systems.

Furthermore, the space within and between atoms is not to be understood as a completely inactive void. Electromagnetic fields and other physical fields play an essential role in describing microscopic systems.

This gives rise to the speculative question of whether an as-yet insufficiently understood relationship might exist between the distribution of energy, existing fields, and the observed gravitational acceleration.

The hypothesis does not yet commit to a specific mechanism. In particular, it is not proven that ordinary electromagnetism can explain gravity. Such an assumption would, among other things, have to explain why gravitational acceleration occurs universally even for electrically neutral bodies.

5. Spacetime Curvature and Alternative Descriptions

GR describes gravity using spacetime geometry and provides numerous experimentally confirmed predictions. The present consideration does not fundamentally challenge these successes.

Rather, it raises the question of whether the geometric description might be a mathematically successful representation of a physical process whose underlying cause is structured differently.

In particular, it must be investigated whether an acceleration generated by energy distribution could produce the same observable effects that are described in GR by spacetime curvature.

The existence of spatially varying accelerations and resulting tidal effects does not automatically rule out an alternative description. However, a corresponding model would have to show how these effects arise and whether it can reproduce the remaining verified predictions of GR.

6. Open Questions and Limits

The present extension is a conceptual consideration and not an elaborated alternative theory of gravity.

In particular, the following questions remain open:

- Which physical mechanism generates the assumed acceleration?
- How does its strength depend quantitatively on energy and its spatial distribution?
- Why does acceleration act independently of the electric charge and material composition of a body?
- Is there a link to electromagnetism, other fields, or a more fundamental physical structure?
- How can tidal effects, light deflection, gravitational time differences, and other confirmed results of GR be described within such a model?

These questions cannot be conclusively answered by this conceptual consideration. Rather, they mark the points where a mathematical elaboration and a comparison with experimental observations would be required.

7. Summary

The extension traces the core assumption "Universe = E" back to the question of the origin of gravity.

It investigates the possibility that observed gravitational acceleration does not necessarily have to be understood as a fundamentally geometric process, but could potentially emerge from a physical effect of spatial energy distribution.

A connection to electromagnetic fields is regarded as a possible, as-yet unresolved direction, not as a proven explanation.

The Einstein field equations are not altered by this consideration. Rather, the extension raises the further question of whether the geometric description of GR might be compatible with a deeper physical interpretation of energy distribution.

The hypothesis remains speculative. Its goal is to formulate an alternative question whose physical viability can only be assessed through a precise mathematical description and comparison with established observations.

Supplementary Note: The Role of Space and a Universal Field

Although the universe is described as an electromagnetic field within this hypothesis, it remains an open question whether such a field permeating all of space is actually a necessary prerequisite for the interactions described.

The considerations regarding gravity raise the further question of whether the observed effects could potentially be explained entirely by the distribution of energy and the interactions between energy states, with matter understood as bound energy. In this case, space itself would not necessarily have to be regarded as an independent physical field.

A universal electromagnetic field therefore remains a possible interpretation, but within the current hypothesis, it is not an established necessity. Whether the interactions can arise entirely from the properties and distribution of energy, or whether a field permeating space is required, remains an open question that warrants further physical investigation.